

Week 3 Introduction

Mathematics for Multidimensional Data

✓ **Video:** 3.1 Multidimensional Data Points and Features
2 min

✓ **Reading:** Multidimensional Data Points and Features Recap
10 min

✓ **Practice Quiz:** Multidimensional Data Points and Features – Review Information
1 question

▶ **Video:** 3.2 Multidimensional Mean
2 min

✓ **Reading:** Multidimensional Mean Recap
10 min

✓ **Practice Quiz:** Multidimensional Mean – Review Information
1 question

▶ **Video:** 3.3 Dispersion: Multidimensional Variables
3 min

✓ **Reading:** Multidimensional Variables Recap
10 min

✓ **Practice Quiz:** Dispersion: Multidimensional Variables – Review Information
3 questions

✓ **Video:** 3.4 Distance Metrics
5 min

✓ **Reading:** Distance Metrics Recap
10 min

📄 **Practice Quiz:** Distance Metrics – Review Information
4 questions



Distance Metrics Recap

Distance metrics

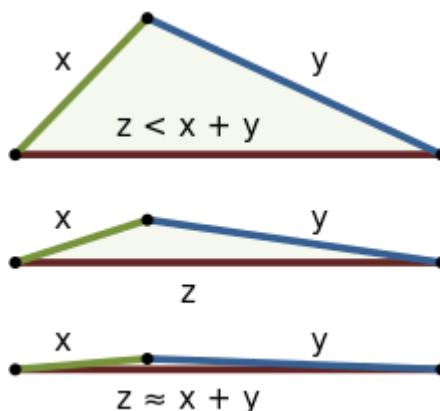
A **distance metric** is what we use to calculate nearness. It is an abstract generalisation of the concept of distance we use every day. The three properties required for a function on a set X defined as $d : X \times X \rightarrow \mathbb{R}$ to be called a **distance function**, are the following, for all $x, y \in X$:

- $d(x, y) \geq 0$, or **non-negativity property**;
- $d(x, y) = 0$ if and only if $x = y$;
- $d(x, y) = d(y, x)$, or **symmetry**.

A distance function is a **metric** if, in addition to the above, it satisfies the following formula:

- $d(x, y) + d(y, z) \geq d(x, z)$, called the **triangle inequality**.

Intuitively, you can see this as saying that the distance function *is the shortest path* between two points. In this case, if you take the distance between x and z , it will be the shortest path, namely: if to go from x to z you pass through another point, y , and you add the two distances from x to y and from y to z , then it can either be the same distance, or greater. You can read more about it [here](#). Here is a visual representation of this (from that same Wikipedia page):



Finally, when you take your set X and consider it together with the distance function d , that pair (X, d) is called a **metric space**.

Types of Distances

The big equation we encountered was this one (Limit the $\sum^m$